Zhenghe Guo

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styokuok.github.io/main GitHub Google Scholar

Final-year undergraduate researcher working on embodied systems and spatial intelligence, with a focus on learned state representations, world modeling, and planning.

Education

Zhejiang University Hangzhou, China
B.Eng. in Agricultural Engineering; Minor in Advanced Class of Engineering Education *2023–2027*
GPA: 3.86/4.00. Undergraduate research with Prof. Mingchuan Zhou.

National University of Singapore Singapore
Visiting Student, Statistics and Data Science *Jun–Jul 2025*
GPA: 4.00/4.00; decision-tree methods instructed by Prof. Wei-Yin Loh.

Research Experience

Westlake University Hangzhou, China
Visiting Research Intern, Inception3D, led by Prof. Anpei Chen *May–Oct 2026*
Research on deformable-object manipulation and physics-grounded world models, combining MPM-based simulation with learned residual dynamics.

Tsinghua University, Institute for AI Industry Research (AIR) Beijing, China
Visiting Research Intern, DISCOVER Lab *Jan–Feb 2026*
Integrated a StereoHO-based perception baseline, calibration, visualization, and robot-side infrastructure for stereo hand-object reconstruction and handover experiments.

Zhejiang University, Robotic Micro/Nano Manipulation Lab Hangzhou, China
Undergraduate Research Assistant, advised by Prof. Mingchuan Zhou *Feb 2025–present*
Developing modeling, simulation, kinematics, and autonomous navigation methods for a magnetically controlled zebrafish robot.

Zhejiang University, State Key Laboratory of CAD&CG Hangzhou, China
Undergraduate Research Assistant, advised by Prof. Ruizhen Hu *Oct 2025–Jun 2026*
Research on instance-level reconstruction, with additional work on physics-grounded articulated-object reconstruction and street-scene reconstruction using COLMAP and FreeTimeGS.

Zhejiang University, Agile Robotic Tele-systems (ARTs) Lab Hangzhou, China
Research Assistant, advised by Prof. Gaofeng Li *Oct 2024–present*
Research on tactile sensing for biomimetic soft-tissue stress-field reconstruction, including sensor-layout optimization, active and sparse sampling, Gaussian-process inference, curved-surface tactile sensing, simulation, and data processing.

Publications

Zhenghe Guo. "ObjectTwin: Measuring the Asset Correspondence Gap for Capture-Anchored Digital Twins." Submitted. [Project page]

Fanghao Wang*, **Zhenghe Guo***, Yihao Hu*, et al. "Spatio-Temporal Transformer Meta-RL for Robust Magnetic Microrobot Navigation in Cluttered Environments." Submitted.

Zhenghe Guo. "Sample-Efficient Residual Dynamics for Contact-Mismatched Deformable Simulation." Submitted. [Project page]

Zhenghe Guo. "Beyond State Prediction: Learning Action Consequences in Visual Physical Systems." Submitted. [Project page]

X. Yang, **Z. Guo**, S. Liang, Z. Yang, C. Zhu, and Z. Zhang. "Rate Maximization for UAV-assisted ISAC System with Fluid Antennas." *2025 IEEE/CIC International Conference on Communications in China Workshops*, pp. 1–5, 2025. [Paper]

Zhenghe Guo et al. "Action Identification and Validation Design for Prediction with Missing Covariates." Submitted to *Journal of Statistical Planning and Inference*. Manuscript available upon request.

*Equal contribution.

Patents

Xiaoyan Chen, Xinyi Fang, Gaofeng Li, Qianming Chen, Zhichao Fan, **Zhenghe Guo**, Peng Deng, Misi Si, Rui Zhang. "Orthodontic Periodontal Ligament Stress Measurement Device and Stress Field Reconstruction Method." Chinese invention patent application. *Pending.*

Gaofeng Li, Zhichao Fan, Xiaoyan Chen, **Zhenghe Guo**, Xinyi Fang, Peng Cheng, Jiming Chen. "Method for Forming and Assembling a Simulated Periodontal Ligament Layer and Associated Testing Device." Chinese invention patent application. *Pending.*

Gaofeng Li, **Zhenghe Guo** (first student inventor), Xiaoyan Chen, Zhichao Fan, Xinyi Fang, Peng Cheng, Jiming Chen. "Physical Field Reconstruction Method and Device." Chinese invention patent application. *Pending.*

Selected Projects

ZJUDancer RoboCup Humanoid Vision and Localization [Team] 2025–2026

- Developed and debugged ball, field, and robot perception, self-localization, and odometry integration for autonomous humanoid soccer; National First Prize and Runner-up at RoboCup China Open 2026.

DYNAMO – Unitree G1 Dance Motion Planning [Project page] 2025

- Designed a dynamics-aware framework combining contact scheduling, ZMP-aware centroidal control, whole-body coordination, and electrical-energy analysis; Asia and Pacific Mathematical Contest in Modeling (APMCM) National First Prize (top 2%).

RoboARM – Vision-Guided Gomoku Manipulation [Code] 2025

- Integrated YOLO and ArUco perception, an HTTP-to-ROS decision bridge, Negamax search, calibration, CAN-based six-axis arm control, and a pneumatic end effector into an autonomous pick-and-place loop.

TurtleBot Autonomous Navigation [Project page] 2025

- Implemented a ROS RRT* global planner with goal-biased sampling, KD-tree collision queries, obstacle inflation, parent selection, rewiring, path publication, and real-robot visualization.

Hybrid QUBO LDPC Decoder [Project page] 2025

- Engineered an LDPC simulation and decoding prototype with belief-propagation baselines, QUBO construction, adaptive penalties, partitioned solving, and BER/FER/runtime evaluation.

Selected Awards

Robotics

- RoboCup China Open, Humanoid League (Small) – National First Prize, Runner-up, Team ZJUDancer, 2026.

Communication and Innovation

- Quantum Information Technology and Application Innovation Contest – National First Prize, Runner-up, 2025.
- China International College Students' Innovation Competition – Gold Medal, 2026.

Mathematical Modeling

- Asia and Pacific Mathematical Contest in Modeling (APMCM) – National First Prize (top 2%), 2025.
- DigitalCup Undergraduate Mathematical Contest in Modeling – National First Prize in Mathematics and Non-Mathematics groups, 2025.
- National Undergraduate Mathematics Competition, Non-Mathematics Category A – Second Prize.

Scholarship

- Xiangyang Scholarship, First Class (¥10,000).

Technical Skills

Programming	Python, C/C++, MATLAB, R, JavaScript/TypeScript, Git, Docker, Linux
Robotics	ROS, OpenCV, PyTorch, CUDA, calibration, kinematics and dynamics, path planning, embedded systems, Arduino/STM32/Raspberry Pi
Perception and ML	3D reconstruction, SfM, visual odometry, object detection, sensor fusion, reinforcement learning, optimization

Languages

Chinese (native) English (TOEFL 110)